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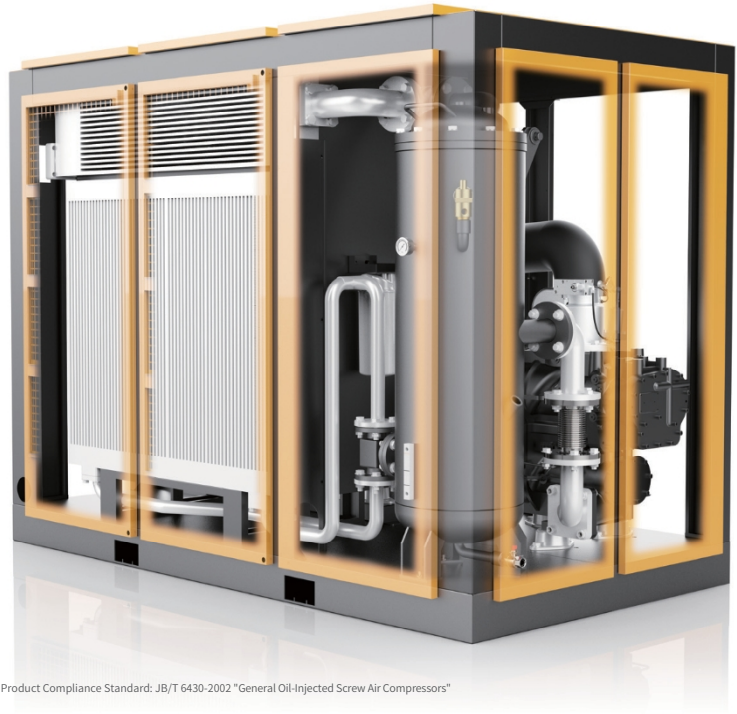
INSTRUCTION MANUAL

User Manual

Permanent Magnet Variable Frequency / Fixed Frequency Screw Air Compressor



Product Compliance Standard: JB/T 6430-2002 "General Oil-Injected Screw Air Compressors"

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Thank you for choosing our screw air compressor! We reserve the right to make design changes to our products and are not responsible for modifying or improving products that have already been shipped. Product specifications and components may be modified in the future without prior notice. All products undergo rigorous inspection and testing before shipment. To ensure safe, long-term operation and optimal performance, operators must read this user manual thoroughly before starting the unit. Do not arbitrarily adjust parameters that have been pre-set at the factory.

Air compressor technology is constantly evolving. We guarantee that the product you have purchased is technologically advanced and represents a superior, energy-efficient option compared to similar models. We value your feedback; please contact us with any suggestions.

1. Transportation and Handling

1. Loading and Unloading

① We recommend using an appropriate forklift for loading and unloading. For products rated at 75 kW or below, please use a 3-ton forklift.For products ranging from 90 kW to 132 kW, please use a 5-ton forklift; for products ranging from 160 kW to 250 kW, please use a 10-ton forklift. An overhead crane may also be used for lifting. When using an overhead crane, insert suitable channel steel into the forklift slots and use a sling spreader to separate the lifting slings from the equipment housing.

警告：不使用撑开器起吊将可能严重损坏设备！

② During loading and unloading, align the forks with the forklift slots on the bottom of the equipment and ensure the forks are long enough.

③ Handle the equipment with care during loading and unloading.

2. Transportation:

This product is precision machinery. We have provided comprehensive protection and packaging, but please still note the following:

① Avoid any collisions during transport to prevent damage to the equipment housing.

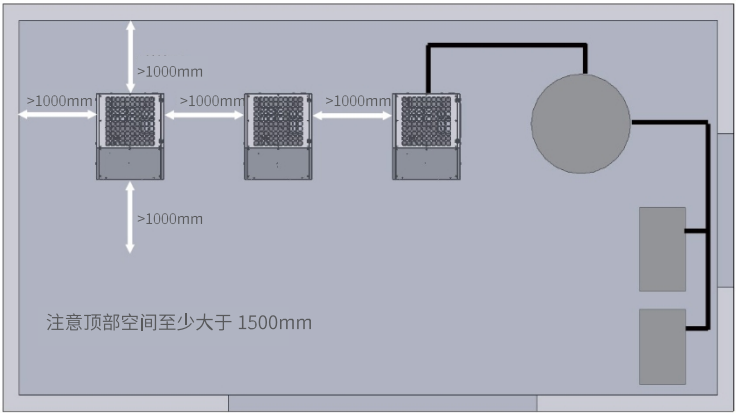
② Do not stack the product during transport, as this may cause the housing to deform.

③ Take measures to protect against moisture during transport; severe humidity may affect the equipment's performance.

2. Installation

1. Site Preparation

Ensure adequate ventilation in the room; otherwise, the compressor may overheat.



2. Cable Preparation

We recommend that you carefully review the IEC 60364-5-523 standard for cable current-carrying capacity before installing cables. We recommend using YJV—copper-core, cross-linked polyvinyl chloride (XLPE)-insulated, polyvinyl chloride (PVC)-sheathed cable. Please select the appropriate cable based on the model of the air compressor you have purchased, as shown in the table below

Compressor Model HP/kW	Voltage (V)	Frequency (Hz)	Maximum Current (A)	Wire Size (mm^2)
7.5/5.5	380	50	15	6
10/7.5	380	50	20	6
15/11	380	50	29	10
20/15	380	50	35	10
30/22	380	50	50	16
40/30	380	50	67	25
50/37	380	50	86	35
75/55	380	50	128	60
100/75	380	50	165	80
125/90	380	50	179	125
150/110	380	50	217	150
175/132	380	50	242	185
220/160	380	50	292	240
270/200	380	50	348	300
340/250	380	50	433	400
430/315	380	50	544	400

3. Selection of Pipe Sizes

When gas flows through a pipe, frictional resistance occurs in straight pipe sections; local resistance occurs at valves, tees, elbows, reducers, and other fittings, resulting in gas pressure loss. The pressure drop over a given pipe length can be calculated using the table below:

Air Flow Rate — Pipe Pressure Drop 2 kg/cm² –(100 m)

Air Flow Rate m^3/min	Pipe Diameter (inches)							
	1/2	3/4	1	1 1/2	1 3/4	2	2 1/2	3
0.8	5.87	1.23	0.339	0.858	0.038			
1.0	9.18	1.92	0.53	0.134	0.059	0.0157		
1.6	23.5	4.9	1.36	0.343	0.152	0.0428		
1.8	62.0	6.20	1.72	0.434	0.192	0.0541		
2.0		7.66	2.12	0.536	0.237	0.0668	0.0177	
2.2		9.26	2.56	0.649	0.287	0.0808	0.0214	
2.4		11.0	3.15	0.772	0.342	0.0967	0.0255	
2.6		12.9	3.58	0.906	0.401	0.113	0.0299	
3.0		17.2	4.77	1.21	0.533	0.150	0.0398	0.0168

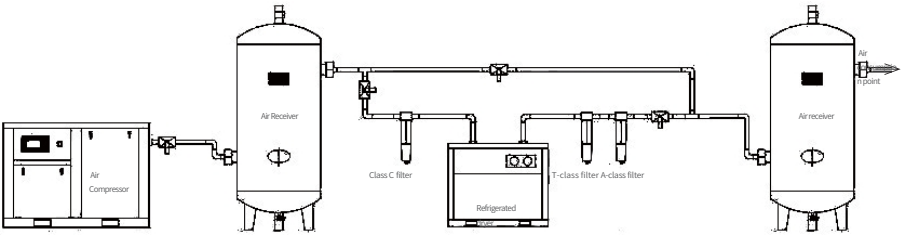
Air flow rate m^3/min	Pipe diameter (inches)							
	3/4	1	1 1/2	1 3/4	2	2 1/2	3	3 1/2
3.4	22	6.13	1.55	0.684	0.193	0.0511	0.0203	
3.8		7.66	1.94	0.855	0.241	0.0636	0.0264	
4.2		9.35	2.36	1.05	0.295	0.0780	0.0311	
4.4		10.3	2.54	1.15	0.323	0.0855	0.0341	0.0158
4.8		12.2	3.09	1.36	0.385	0.102	0.0406	0.0189
5.0		13.3	3.35	1.48	0.418	0.111	0.044	0.0204
5.5		16.0	4.06	1.79	0.505	0.134	0.0533	0.0244
6.0		19.1	4.82	2.13	0.601	0.159	0.0634	0.0299

Air Flow Rate m^3/min	Pipe Diameter (INCH)						
	1 1/2	2	2 1/2	3	3 1/2	4	5
6.5	2.50	0.709	0.187	0.0744	0.0346		
7.0	2.90	0.815	0.217	0.0863	0.0401		
7.5	3.33	0.940	0.249	0.0990	0.0460		
8.0	3.79	1.0	0.283	0.113	0.0524	0.0247	
8.5	4.28	1.21	0.319	0.127	0.0590	0.0309	
9.0	4.80	1.35	0.358	0.143	0.0662	0.0347	
9.5	5.35	1.51	0.399	0.159	0.0738	0.0386	
10	5.93	1.67	0.442	0.176	0.0818	0.0428	
11	7.17	2.02	0.535	0.203	0.0990	0.0518	0.0165
12	8.53	2.40	0.637	0.254	0.118	0.0616	0.0196
13	10.0	2.82	0.747	0.298	0.138	0.0723	0.0230
14	11.6	3.76	0.995	0.396	0.184	0.0963	0.0306

Note:

1. Actual pressure drop in a straight pipe section = Table value × Pipe length / (100 × Compression ratio). (Compression ratio = Gauge pressure + 1)
2. The total pressure drop in the piping system must also include local pressure losses caused by elbows, reducers, tees, valves, etc. These values can be found in relevant handbooks.

We recommend referring to the diagram below for system configuration:



3. Safety Precautions

1. Legend (Caution symbol) (Warning symbol) (Prohibition symbol)

We take safety very seriously; please be sure to read the safety precautions in this section carefully before use.

Electrical Safety

1. Select the correct power cord gauge based on the air compressor's power rating. Do not use a cord with an insufficient gauge, as this may cause the cord to overheat and burn out, posing a safety hazard.
2. It is best to use a dedicated power system for the air compressor, and in particular, avoid connecting it in parallel with other electrical loads. If used in parallel, excessive voltage drops or three-phase current imbalance may cause the air compressor to overload, triggering the protective device to trip and affecting other electrical equipment; this is especially important for high-power air compressors.
3. The operating current of the motor must not exceed 15% of the rated current. If there is an imbalance in the three-phase currents, the difference between the lowest and highest phase currents must not exceed 5%. If there is a voltage drop in the power supply, the voltage drop must not be less than 5% of the rated voltage.
4. When distributing power, ensure that the supply voltage matches the motor's rated voltage.
5. A grounding wire must be installed for the motor or system to prevent hazards caused by electrical leakage. Note: The grounding wire must not be connected directly to the air delivery pipe or cooling water pipe.
6. Select and install an appropriate circuit breaker based on the air compressor's power rating to protect the system's electrical switches and ensure safety.

Installation Process

General Precautions

1. Operators must follow safety operating procedures and comply with all relevant workplace safety requirements and regulations.
2. If any of the following instructions conflict with applicable regulations, the stricter of the two shall prevail.
3. Installation, operation, maintenance, and repair work must only be performed by authorized, trained, and qualified professionals.
4. The air produced by the compressor is not considered to be of breathing quality. To ensure the compressed air meets breathing quality standards, it must be adequately purified in accordance with applicable regulations and standards.
5. Before performing any maintenance, repairs, adjustments, or other non-routine inspections, stop the compressor, press the emergency stop button, disconnect the power supply, and relieve the pressure in the compressor. In addition, the power disconnect switch must be opened and locked out.
6. Do not play with compressed air. Do not allow the air to come into contact with your skin or direct the airflow at people. Do not use compressed air to remove dust from clothing. When using compressed air to clean equipment, exercise caution and wear safety goggles.
7. The user is responsible for maintaining the unit in a safe operating condition. If parts or accessories are unsuitable for safe operation, they must be replaced.

Precautions During Installation

1. This machine must only be lifted using appropriate equipment in accordance with applicable safety regulations. Before lifting, all loose or rotating parts must be securely fastened. It is strictly prohibited to pause or linger in hazardous areas while lifting heavy loads. Acceleration and deceleration during lifting must remain within safe limits. A safety helmet must be worn when working at heights or in areas where lifting equipment is in use.
2. Place the machine in an environment with cool, clean air. If necessary, install an air intake duct. Do not block the air inlet. Measures must be taken to minimize moisture in the incoming air as much as possible.
3. Before connecting the piping, all blind flanges, plugs, caps, and desiccant bags must be removed.
4. Air hoses must be of the correct size and suitable for use at operating pressure. Do not use worn, damaged, or old hoses. Branch pipes and fittings must be of the correct size and suitable for use at operating pressure.
5. The intake air must not contain flammable gases, vapors, or particles (e.g., paint solvents) that could cause an internal fire or explosion.
6. Position the intake opening to prevent the suction of loose clothing worn by operators.
7. Ensure that the discharge line from the compressor to the aftercooler or air distribution network can expand freely when heated and is not near or in contact with flammable materials.
8. Do not apply external force to the air outlet valve; do not kink or deform the connecting hose.
9. If a remote control is installed, the machine must be clearly marked with the following warning: Danger: This machine is controlled remotely and may start without warning. Before performing maintenance or repairs, the operator must ensure that the machine has been shut down and that the isolating switch has been opened and locked. To further ensure safety, the person turning on the power to the remotely controlled machine should take sufficient precautions to ensure that no one is inspecting or operating the machine. Therefore, a corresponding notice should be posted on the starting equipment.
10. Air-cooled machines must be installed in such a way that sufficient cooling air is available and that exhaust air does not recirculate into the compressor air inlet or the cooling air inlet.
11. Electrical connections must comply with applicable codes. The machine must be grounded, and fuses must be installed in each phase to prevent short circuits. A lockable power disconnect switch must be installed near the compressor.
12. On machines equipped with an automatic start-stop system or with the automatic restart function activated after a power failure, a label must be affixed near the instrument panel stating, “This machine may start without warning.”
13. In multi-compressor systems, manual valves must be installed to isolate each compressor. Check valves must not be used to isolate the pressure system.
14. Do not remove or tamper with the safety devices, protective devices, or insulators installed on the machine. One or more pressure relief devices must be used as necessary to protect each pressure vessel or auxiliary equipment installed externally on the machine that is pressurized above atmospheric pressure.
15. Pipes or other parts that exceed 80°C (176°F) and may be accidentally touched by personnel during normal operation must be protected or isolated; other high-temperature pipes must be clearly marked.
16. For water-cooled machines, the cooling water system installed outside the machine must be protected by a safety device set to the pressure of the main cooling water inlet.
17. If the floor is uneven or sloped, consult the manufacturer

Safety Measures During Operation General Precautions

1. Operators must follow safe operating procedures and comply with all relevant occupational safety requirements and regulations.
2. If any of the following instructions conflict with applicable regulations, the stricter of the two shall prevail.
3. Installation, operation, maintenance, and repair work must only be performed by authorized, trained professionals.
4. The air produced by the compressor is not considered suitable for breathing. To make the compressed air suitable for breathing, it must be adequately purified in accordance with applicable regulations and standards.

5. Before performing any maintenance, repair, adjustment, or other non-routine inspection, stop the compressor, press the emergency stop button, disconnect the power supply, and depressurize the compressor. Additionally, the power isolation switch must be opened and locked out.

6. Do not play with compressed air. Do not allow the air to come into contact with your skin or direct the airflow at people. Do not use compressed air to remove dust from clothing. When using compressed air to clean equipment, exercise caution and wear safety goggles.

Precautions During Operation

1. Use only hoses, fittings, and connections of the correct type and size. When venting through a hose or air line, ensure the open end is securely held in place. If the open end is left unsecured, it may move suddenly and cause injury. Before disconnecting a hose, ensure it is completely depressurized.
2. The person turning on the power to the remotely controlled machine must take sufficient precautions to ensure that no one is inspecting or operating the machine. Therefore, a corresponding notice should be posted on the remote start device.
3. Do not operate the machine where flammable or toxic gases, vapors, or particles may be inhaled.
4. Do not operate the machine below or above the rated limits.
5. Keep all cabinet doors closed during operation. These doors may only be opened briefly when performing routine inspections or similar operations. Wear ear protection when opening the machine doors.
6. Persons in environments or rooms where the sound pressure level reaches or exceeds 90 dB(A) should wear ear protection.
7. Please check regularly: · All protective devices are in place and securely fastened. · All hoses and/or pipes inside the machine are in good condition, safe, and free of wear. · There are no leaks. · All fasteners are tightened · All electrical leads are secure and neatly arranged · Safety valves and other pressure relief devices are not blocked by dirt or paint · Air outlet valves and the air distribution system (i.e., pipes, couplings, manifolds, valves, hoses, etc.) are well-maintained and show no signs of wear or misuse
8. If the hot air discharged from the compressor is used for an air heating system (e.g., to heat a workroom), take precautions to prevent air contamination and potential contamination of breathing air.
9. Do not remove or tamper with any sound-insulating material.
10. Do not remove or tamper with any safety devices, protective devices, or insulation installed on the machine. One or more pressure relief devices shall be used as necessary to protect each pressure vessel or auxiliary equipment installed outside the machine that is pressurized above atmospheric pressure.

Safety Measures During Maintenance or Repair General Precautions

1. Operators must follow safe operating procedures and comply with all applicable local occupational safety requirements and regulations.
2. If any of the following instructions conflict with applicable regulations, the stricter of the two shall prevail.
3. Installation, operation, maintenance, and repair work must only be performed by authorized, trained, and qualified personnel.
4. Air produced by the compressor is not considered to be of breathing quality. To make the compressed air suitable for breathing, it must be adequately purified in accordance with local regulations and standards.
5. Before performing any maintenance, repair, adjustment, or other non-routine inspection, stop the compressor, press the emergency stop button, disconnect the power supply, and depressurize the compressor. In addition, the power isolation switch must be opened and locked.
6. Do not play with compressed air. Do not allow the air to come into contact with your skin or direct the airflow at people. Do not use compressed air to remove dust from clothing. When using compressed air to clean equipment, exercise caution and wear safety goggles.

Precautions During Maintenance or Repair

1. Always wear safety goggles.

2. Use only the correct tools when performing maintenance and repair work.

3. Use only genuine spare parts.

4. All maintenance work should only be performed when the machine has cooled down.

5. A warning sign with text such as “Work in progress; do not start” should be affixed to the starting equipment.

6. The person turning on the machine’s power via remote control must take adequate precautions to ensure that no one is inspecting or operating the machine. Therefore, a corresponding notice should be posted on the remote start device.

7. Close the compressor air outlet valve before connecting or disconnecting pipes.

8. Before disassembling any pressurized components, effectively isolate the machine from all pressure sources and relieve the entire pressure system.

9. Do not use flammable solvents or carbon tetrachloride to clean parts. Take safety precautions against toxic fumes emitted by cleaning fluids.

10. During maintenance and repair, carefully check the cleanliness of the machine. Cover parts and openings with a clean cloth, paper, or tape to prevent dust accumulation.

11. Do not perform welding or any other heat-generating operations near the lubrication system. Before performing such operations, the oil tank must be thoroughly cleaned, for example, by steam cleaning. Do not weld on pressure vessels or modify them in any way.

12. Whenever there are signs or suspicion that an internal component of the machine is overheating, stop the machine immediately; however, do not open the inspection cover until sufficient cooling time has elapsed to avoid the risk of spontaneous combustion of oil vapors when air enters.

13. Do not use open flames to inspect the interior of the machine, pressure vessels, etc.

14. Ensure that no tools, loose parts, or rags are left inside or on the machine.

15. All adjustment and safety devices should be serviced regularly to ensure they function properly. These devices must not fail.

16. After maintenance or major repairs, check that the operating pressure, temperature, and time settings are correct before cleaning the machine for use. Check that all controls and shut-off devices are installed and functioning properly. If the coupling guard on the compressor drive shaft has been removed, check that it has been reinstalled.

17. Each time the separator filter element is replaced, check for carbon buildup in the discharge pipe and inside the oil-air separator tank; if there is excessive carbon buildup, it should be removed.

18. Protect the motor, air filter, electronic components, and control elements from moisture ingress, for example, during steam cleaning.

19. Ensure that all sound-absorbing materials and vibration dampers (e.g., those on the housing and in the compressor’s air inlet and outlet systems) are in good condition. If damaged, replace them with original materials provided by the manufacturer to prevent an increase in sound pressure level.

20. Do not use corrosive solvents that could damage air ducting materials (e.g., polycarbonate bases).

21. When handling refrigerants, pay special attention to the following safety precautions:

- Do not inhale refrigerant vapors. Ensure the work area is well-ventilated; use respiratory protection if necessary.
- Always wear protective gloves. If refrigerant comes into contact with the skin, rinse the skin with water. If liquid refrigerant penetrates clothing and comes into contact with the skin, do not hastily remove or take off the clothing; instead, rinse the clothing with plenty of fresh water until all refrigerant is washed away, then seek immediate medical attention.

22. Protect your hands to prevent injury from contact with hot machine parts, such as during the oil drainage process.

4. Operating Procedures

Initial Installation:

1. Ensure that the inlet air temperature or ambient temperature is not below 0°C. If it is below 0°C, contact the manufacturer for a solution.

2. Remove the transport securing bolts from the main unit, motor, and oil separator tank; these bolts are typically painted red at the top. (If the equipment needs to be transported again, these transport securing bolts must be reinstalled.)

3. Check that the electrical connections are correct and that the cable terminals are secure and reliable.

4. Check that the air line connections are secure.

5. Before commissioning, ensure that no one is present in the hazardous areas near the motor or compressor.

6. When operating the screw compressor, the doors and panels must be closed.

7. If the screw compressor is being connected to a power source for the first time, check the rotation direction of the drive motor. If the rotation direction is incorrect, immediately press the emergency stop button to shut down the machine. Otherwise, the compressor may suffer severe damage. You may temporarily open the machine door to check the rotation direction.

8. After connecting the main power supply, all indicator lights and segments on the LCD display will illuminate to perform a display test.

9. The screw compressor is equipped with an automatic operation monitoring system that will shut down the machine in the event of a malfunction.

10. Press the compressor start button.

5. Control Panel Instructions

Due to design requirements or stability considerations, this product may use control panels manufactured by different vendors. Operating procedures may vary by manufacturer. A control panel operation manual and inverter manual are provided with this product; please refer to them.

6. Troubleshooting

Problem	Possible Cause	Troubleshooting Method
1. Fails to Start	1. Blown fuse 2. Protective relay tripped 3. Poor contact in the start button 4. Voltage too low 5. Motor malfunction 6. Unit malfunction	1. Have an electrician inspect and replace 2. Have an electrician inspect and replace 3. Have an electrician inspect and replace 4. Have an electrician inspect and replace 5. Have an electrician inspect and replace 6. Manually rotate the unit; if it cannot be turned, contact the manufacturer
2. High exhaust temperature (shut down if it exceeds 105°C)	1. Ambient temperature is too high 2. Thermostatic valve malfunction 3. Insufficient lubricating oil 4. Oil cooler fins are excessively dirty 5. Clogged oil filter 6. Cooling fan malfunction 7. Temperature sensor failure	1. Improve ventilation 2. Check/replace the thermostat 3. Check / adjust oil level 4. Clean the radiator fins 5. Replace the oil filter 6. Replace the cooling fan 7. Check / Replace Temperature Sensor
III. Low Exhaust Pressure	1. Air demand exceeds supply 2. Clogged air filter 3. Intake valve not opening fully 4. Clogged oil-air separator element	1. Check for leaks in the piping 2. Clean or replace the filter element 3. Check the intake valve 4. Check the pressure readings on the pressure gauges before and after the oil-gas separator element and on the LCD display; replace if necessary
4. Shutdown due to excessive exhaust pressure	1. Pressure transmitter malfunction or abnormal interference signal 2. Exhaust valve is closed	1. Troubleshoot or replace 2. Open the exhaust valve
5. Unable to idle; system pressure remains at operating pressure or continues to rise during idling	1. Intake valve malfunction 2. Pressure sensor failure 3. Relief valve failure	1. Check the intake valve 2. Repair; replace if necessary 3. Repair; replace if necessary
6. Frequent empty and loaded trips	1. Pipe Leaks 2. Pressure differential between empty and loaded vehicles set too low 3. Unstable air consumption	1. Inspect the piping 2. Reset . 3. Increase air tank capacity

VII. High oil content in the air, excessive oil consumption	1. Oil level too high 2. Clogged return line filter or throttle orifice 3. Damaged oil-air separator element or gasket 4. Leak in the lubrication system 5. Exhaust pressure too low 6. Excessive oil foaming	1. Check the oil level and drain oil to the normal level 2. Clean the filter screen and throttle orifice of the return line filter; replace if necessary 3. Inspect the filter element and gasket; replace if damaged 4. Inspect the piping 5. Increase the vent pressure 6. Change the lubricating oil
8. Oil mist escapes from the air filter when the machine is shut down	1. Intake valve is not sealing properly 2. Shutdown under heavy load 3. Minimum pressure valve leak 4. Damaged oil-fine separator	1. Check 2. Check if the intake valve is fully closed 3. Repair; replace if necessary 4. Replace

7. Maintenance and Care

7.1 Safety Precautions for Maintenance

Before performing maintenance, repairs, or adjustments, please do the following:

- Stop the compressor.
- Close the air outlet valve,
- Press the emergency stop button.
- Disconnect the power supply. Hang a warning sign on the power switch.
- Depressurize the compressor.
- Close the shut-off valve connecting the air compressor to the supply system to prevent compressed air from flowing back into the section under maintenance. Never rely on the air compressor's check valve to isolate the supply system.
- Poor maintenance not only affects the normal operation of the unit but may also compromise operator safety.
- Do not remove nuts, oil plugs, or other parts while the air compressor is running or under pressure.
- Do not use flammable solvents, such as gasoline or kerosene, to clean air filters or other components. Use safe solvents as specified in the instructions.
- For water-cooled units, the water supply system must be shut off and the pressure in the water lines released. Ensure the air compressor unit has cooled down to prevent scalding or burns. Wipe up oil and water spills from the floor to prevent slipping.

7.2 Product Warranty Obligations

Use only authorized and approved parts. Any damage or malfunction resulting from the use of unauthorized or unapproved parts is not covered by the warranty or product liability.

7.3 Maintenance Intervals

Develop a maintenance schedule based on the compressor's environment and operating conditions, particularly regarding maintenance intervals.

In addition to daily, weekly, monthly, and quarterly inspections, the schedule below specifies other preventive maintenance operations.

Each maintenance plan has a set interval, and all maintenance operations included in that plan must be performed according to this interval. When the interval is reached, a message will appear on the screen indicating which maintenance plan is to be executed; the interval must be reset after the maintenance operation is completed;

7.4 General Maintenance Schedule

Checklist

Cycle	Operation
Daily	a. Check the oil level. Check the reading on the display. b. For water-cooled units: Check the cooling water flow.
Weekly	a. Check the unit for unusual noises and leaks; b. Check that the instrument readings are correct; c. Check that the temperature display is functioning normally.
Monthly	a. Inspect the unit for rust or loose parts; if rust is present, remove it and apply oil or paint; tighten any loose parts; b. Drain the condensate.
Every three months	a. Inspect the cooler and clean it if necessary. Remove and inspect the air filter element. Clean it with compressed air if necessary. Replace any damaged or heavily contaminated filter elements. Inspect the pre-filter element (if installed). Replace it if necessary. b. Clean more frequently when operating in dusty environments. c. Inspect hoses for signs of aging or cracking; d. Inspect electrical components and clean the control panel.

Important Notes

- For oil and oil filter replacement intervals under extreme temperature, humidity, or cooling air conditions, please consult the after-sales service center.
- Any leaks should be addressed immediately. Damaged pipes or fittings must be replaced.

7.5 Drive Motor

Bearing Maintenance

Some models require periodic lubrication of the drive motor.

7.6 Oil Specifications

Never mix different brands or types of lubricating oil, as they may be incompatible and the resulting mixture will have inferior properties. A label indicating the type of oil used at the factory is affixed to the air receiver tank or the machine housing. Oil should be drained from the compressor at all drain points on a regular basis. Waste oil remaining inside the compressor will contaminate the lubricant

lubrication system and shorten the life of new oil.

We strongly recommend using our company's genuine, proprietary lubricants. They are the result of our many years of field experience.

7.7 Schematic of Maintenance Disassembly Points

7.8 Maintenance and Care of the Intake Air Filter

The air filter is a component designed to remove dust and contaminants from the air; the filtered, clean air enters the screw compressor's compression chamber for compression.Since the internal clearances of the screw compressor only allow particles smaller than 15 microns to pass through, if the air filter element becomes clogged or damaged, a large amount of particles larger than 15 microns will enter the compressor's internal circulation. This not only significantly shortens the service life of the oil filter and fine oil separator elements but also causes a large amount of particles to enter the bearing chamber directly, accelerating bearing wear, increasing the rotor clearance, reducing compression efficiency, and potentially causing the rotors to seize.

7. The air filter element should be maintained once a week. Unscrew the retaining clip on the cover, remove the air filter element, and use compressed air at 0.2–0.4 MPa to blow dust particles off the outer surface of the air filter element from the inside out. Wipe away any dirt from the inner walls of the air filter housing with a clean cloth.Reinstall the air filter element, ensuring that the sealing ring at the front end of the element is tightly seated against the inner surface of the housing.

8. Under normal conditions, the air filter element should be replaced every 1,000–1,500 hours. In particularly harsh environments, such as mines, ceramic factories, and cotton mills, it is recommended to

500 hours.

9. When cleaning or replacing the air filter element, ensure all components are properly aligned to prevent foreign objects from falling into the intake valve.

10. Regularly inspect the intake hose for damage or fatigue and check the connections between the hose and the air filter intake valve for loose or cracked fittings. If any issues are found, repair or replace them promptly.

7.9 Replacement of the Oil Filter

1. The oil filter element should be replaced after the first 500 hours of operation on a new machine. Use a special wrench to unscrew the oil filter element counterclockwise to remove it. Before installing the new element, it is best to add screw-type compressor oil. To ensure a tight seal, use both hands to screw the oil filter element back into the housing and tighten it firmly.

2. It is recommended to replace the oil filter element every 1,500–2,000 hours. It is best to replace the oil filter element at the same time as changing the oil. In harsh operating environments, the replacement interval should be shortened.

3. It is strictly prohibited to use the oil filter beyond its service life; otherwise, severe clogging of the filter may cause the pressure differential to exceed the bypass valve's tolerance limit, causing the bypass valve to open automatically. This will allow large amounts of contaminants and particles to enter the screw compressor directly with the oil, resulting in serious consequences.

4. The replacement of the diesel engine oil filter and diesel fuel filter in diesel-powered screw compressors should follow the maintenance requirements for the diesel engine; the replacement procedure is similar to that of the screw compressor oil filter.

7.10 Maintenance and Replacement of the Oil-Air Separator

1. The oil-air separator is a component that separates screw compressor lubricating oil from compressed air. Under normal operating conditions, the service life of the oil-air separator is approximately 3,000 hours; however, the quality of the lubricating oil and the filtration precision of the air significantly affect its lifespan. Therefore, in harsh operating environments, the maintenance and replacement cycle for the oil-air separator must be shortened, and the installation of a pre-air filter should even be considered.

2. The fine oil separator must be replaced when it reaches the end of its service life or when the pressure differential exceeds 0.12 MPa. Failure to do so may result in motor overload, damage to the fine oil separator, and oil leakage. 3. Replacement Procedure:

a. Remove all control pipe fittings installed on the oil-gas separator tank cover. Remove the return oil pipe extending from the tank cover into the tank, and unscrew the fastening bolts on the tank cover.

b. Remove the oil-gas tank cover and take out the oil-gas separator. Remove the asbestos gasket and any debris adhering to the cover plate.

c. Install the new fine oil separator, ensuring it is centered and that the bottom sealing ring makes full contact. When tightening the top screw, take care not to deform the separator core.

d. Reinstall the top cover, return oil pipe, and all control pipes as originally configured, and check for leaks.

7.11 Maintenance and Replacement of Screw Compressor Oil

The quality of screw compressor oil has a decisive impact on the performance of oil-injected screw compressors. High-quality oil features excellent oxidation stability, rapid separation, superior defoaming properties, high viscosity, and effective corrosion protection. Therefore, users must use genuine, company-specific screw compressor oil.

1. Perform the first oil change after the 500-hour break-in period for new machines; thereafter, change the oil every 2,000 operating hours. It is recommended to replace the oil filter at the same time as changing the oil. Shorten the replacement interval when operating in harsh environments.

2. Procedure:

a. Start the air compressor and run it for 5 minutes to raise the oil temperature above 50°C, allowing the oil viscosity to decrease.

b. Stop the unit. When the oil receiver maintains a pressure of 0.1 MPa, open the drain valve at the bottom of the oil receiver and connect a storage tank. Open the drain valve slowly to prevent hot, pressurized lubricating oil from splashing and causing injury or contamination. Close the drain valve once the oil begins to drip. Unscrew the oil filter element, drain all lubricating oil from the piping simultaneously, and install a new oil filter element.

c. Open the filler plug, fill with new oil until the oil level is within the range indicated by the dipstick, tighten the filler plug, and check for leaks.

d. The lubricating oil must be checked frequently during use. If the oil level is too low, top up with fresh oil immediately. Condensate must also be drained regularly; generally once a week, but every 2–3 days in hot weather. If the machine has been shut down for more than 4 hours, open the drain valve when there is no pressure in the oil-gas tank to drain the condensate. Close the valve immediately once oil begins to flow out.

e. Never mix lubricating oils of different brands, and never use lubricating oil past its expiration date. Doing so will degrade the oil's quality, impair its lubricating properties, and lower its flash point, which can easily lead to shutdowns due to overheating and cause the oil to spontaneously combust.

7.12 Periodic Maintenance of the Air Compressor

(1) Every 100 hours or within one week of rig operation: The primary filter screen of the air compressor oil filter and the coarse filter screen on the oil tank must be cleaned. During cleaning, remove the filter element and use a wire brush to remove debris from the screen. Since the air compressor operates in a dusty environment, frequent cleaning of the air filter and oil filter is critical; neglecting this will lead to a series of malfunctions.

(2) Every 300 hours: Clean the air compressor filter element. After cleaning, blow it dry and soak it in the same oil from the tank. If dust levels are severe, adjust the cleaning interval based on actual conditions.When the red line on the air contamination indicator of the unloading valve becomes visible, it indicates that the air filter is severely contaminated and should be cleaned. Clean or inspect components such as the unloading valve, minimum pressure valve, and automatic bleed valve. Thoroughly clean the oil filter, removing all contaminants deposited at the bottom of the oil filter.

(3) Every 1,000 hours: Replace the paper filter element of the air compressor oil filter. Check whether the safety valve, temperature control valve, pressure regulator, and pressure relief valve are functioning normally;

(4) Every 3,000 hours: Replace the oil as needed based on the degree of emulsification, the amount of impurities, and changes in viscosity. The oil-air separator filter element typically lasts 2,000 to 4,000 hours.No special maintenance is required during this period. In dusty environments, dust tends to accumulate on the oil cooler, reducing cooling efficiency. Clean as needed based on specific conditions; generally, cleaning is required once every 4 months to ensure effective heat dissipation; (5) Perform a major overhaul of the main unit every 10,000 hours, replacing radial and thrust bearings as necessary.

The advantage of this unit is that it requires minimal routine maintenance; however, to ensure the compressor unit operates normally and reliably with a long service life, a detailed maintenance plan must be established. Designate specific personnel for operation, and implement regular maintenance and periodic inspections to keep the compressor clean, free of oil, and free of dirt. Please refer to the table below for maintenance of major components.

7.13 Cooler Maintenance

The cooling efficiency of the cooler directly affects the operating temperature of the air compressor. The plate-fin structure is prone to dust accumulation, so the cooler must be maintained by blowing dry compressed air at 0.4 MPa or higher from top to bottom during each shift.For diesel-driven screw compressors, the cooler is mounted vertically. During maintenance, use dry compressed air at 0.4 MPa or higher to blow from the outside inward. After blowing, thoroughly clean dust and particles from the air guide hood to prevent them from being blown back into the cooler by the fan, which could cause blockages and lead to overheating. If the machine is used in harsh environments and the cooler surface is covered with oil residue, it must be cleaned with a carbon deposit remover.

7.14 Scheduled Maintenance Schedule

Item	Description	Inspection or Replacement Interval (Hours)					Remarks
		500	1,000	2000	2,500	4000	
Air Filter Cartridges	Removes surface dust and impurities	-					Extend or shorten the replacement interval based on visible dust accumulation and operating conditions
	Replace with a new filter element				-		
Intake valve piston seal	Inspect or replace the seal	-					
Compressor lubricant	Is there enough?	-					
	Change to new oil				-		
Oil filter	Replace with a new part				-		
Oil-air separator	Replace with new part				-		
Minimum pressure valve	Check opening pressure					-	Clean
Cooler Dust Removal	Remove dust from heat dissipation surfaces	-					
Safety valve	Check for proper operation					-	
Drain valve	Drain water and debris	-					
Motor	Apply grease					-	Follow the motor's operating instructions

Note: A. When performing maintenance or replacing parts as per the table above, ensure that: all pressure within the compressor system has been released; the system has been isolated from other pressure sources; the main circuit breakers

are disconnected, and safety signs prohibiting re-energization have been posted.

B. The replacement interval for compressor cooling lubricating oil depends on the operating environment, humidity, dust levels, and the presence of acidic or alkaline gases in the air. a. Machines using standard cooling oil: Replace the oil after the first 500 hours of operation for a new compressor; thereafter, replace it every 2,500 hours or annually according to the normal oil change cycle. For air compressors not in continuous operation, change the oil as needed. For compressors in storage, change the oil every 2 years.

C. The oil filter must be replaced after the first 500 hours of operation, the second time after 1,000 hours, and thereafter every 2,500 hours according to the standard schedule.

D. When servicing or replacing the air filter or intake valve, take care to prevent any debris from entering the compressor head; seal the main unit's inlet during the operation. After completing the work, manually rotate the main unit several times in the direction of rotation to ensure there are no obstructions before starting the unit.

E. The oil filter must be replaced every time the oil is changed.

e of Conformity

Inspector:

Please read this manual carefully before use. The company reserves the right to interpret this manual. The actual product appearance may differ from the illustrations. Product upgrades may occur without prior notice!